INCH-POUND
MIL-C-16290K(NU)

24 April 1991
SUPERSEDING
MIL-C-16290J(NU)
27 October 1983

MILITARY SPECIFICATION

CLOTH, MELTON, WOOL (MOTH-PROOFED)

This specification is approved for use by the Navy Clothing and Textile Research Facility, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for wool melton cloth.

* 1.2 Classification. The melton cloth shall be of the following types (see 6.2):

Type I - 16 ounce - Blue 3346

Type II - 22 ounce - Blue 3346

2. APPLICABLE DOCUMENTS

* 2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Officer in Charge, Navy Clothing and Textile Research Facility, P.O. Box 59, Natick, MA 01760-0001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8305

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SPECIFICATIONS

FEDERAL

PPP-P-1132 - Packaging of Woolen; Worsted and Wool Blend (Synthetic Fiber; Cotton) Fabrics

MILITARY

MIL-C-43665 - Cloth Wool: Mothproofing Treatment of

STANDARDS

FEDERAL

FED-STD-191 - Textile Test Methods

MILITARY

MIL-STD-655 - Provisions for Evaluating Quality of Cloth, Wool, Worsted and Wool Blends
* MIL-STD-2073-1 - DoD Materials Procedures for Development and Application of Packaging Requirements
* MIL-STD-2073-2 - Packaging Requirements Codes

* (Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standarization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094).

* 2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

U.S. DEPARTMENT OF AGRICULTURE

Methods of Test for Grades of Wool Top

(Applications for copies should be addressed to the U.S. Department of Agriculture, Agricultural Marketing Service, Washington, DC 20250-0001).

RULES AND REGULATIONS UNDER THE WOOL PRODUCTS LABELING ACT of 1939

(Application for copies should be addressed to the Federal Trade Commission, Washington, DC 20580-0001).

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

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TECHNICAL MANUAL OF THE AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS

Method No. 8-1988 - Colorfastness to Crocking: AATCC Crockmeter Method
Method No. 16-1987 - Colorfastness to Light: General Method

(Copies may be obtained from AATCC National Headquarters, Post Office Box 12215, Research Triangle Park, NC 27709-2215).

* (Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services).

* 2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

* 3.1 Standard sample. The finished cloth shall match the standard for shade and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3). The standard sample is identified under piece number 84241.

3.2 First article. When specified the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.3 and 6.2).

3.3 Material.

* 3.3.1 Wool. The stock shall be fleece or pulled wool or both, not lower in grade than 64's U.S. Standard, when tested as specified in 4.4.1.1. The use of noils, laps or any other wool manufacturing by-products is prohibited.

3.3.2 Yarns. The wool stock shall be thoroughly scoured, blended, carded, and evenly spun on the woolen system into singles yarn for both warp and filling.

NOTE: Selvage yarns: To prevent edges of the cloth from building up when rolled, the selvage yarns may be made of a finer count than the body of the cloth. These yarns shall also consist of wool fibers.

3.4 Weave. The weave for type I shall be a 2 up, 1 down right hand twill and the weave for type II shall be a 3 up, 1 down crowfoot when tested as specified in 4.6.

3.5 Physical requirements. The finished cloth shall conform to the requirements shown in Table I when tested as specified in 4.6.

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Table I. - Physical requirements

Type	Ounces per linear yards 1/ (Grams per linear m - 1.72 m wide) Min.	Max.	Yarns per inch (2.54 cm) Minimum Warp	Filling	Breaking Strength in Pounds (Newton's) Minimum Warp	Filling	Shrinkage (percent) Minimum Warp	Filling
I	16(448g)	17(476g)	60	55	58	46	4.0	3.0
II	22(616g)	24(672g)	55	45	80	60	4.5	2.0

1/ Based on 56 inch width.

* 3.6 Width. The width of the cloth shall be as specified (see 6.2), and shall be the minimum acceptable width inclusive of selvage when fly shuttle looms or shuttleless with tuck-in selvage looms are used. For all other shuttleless looms, the width measurement shall be made between the last warp yarn on each side excluding the protruding fringe(s).

3.7 Wool content. The finished cloth shall contain not less than 95.0 percent wool when tested as specified in 4.6.

3.8 pH. The pH value of the water extraction of the finished cloth shall be no less than 4.0 nor more than 8.0 when tested as specified in 4.6.

3.9 Length and put-up. Unless otherwise specified (see 6.2), the cloth shall be furnished in continuous lengths each not less than 40 yards (36.56 m). Each length shall be put-up on full width rolls as specified in PPP-P-1132.

3.10 Mothproofing. Mothproofing of the wool material shall be in accordance with MIL-C-43665.

3.11 Color. The color of the finished cloth shall be Blue 3346 and shall match the standard sample. The color shall be produced by stock dyeing with suitable dyestuffs of required fastness properties to match the standard sample.

* 3.11.1 Matching. The color of the finished cloth shall match the standard sample when viewed under filtered tungsten lamps which approximate artificial daylight having a color temperature of 7500 (± 200) Kelvin, with illumination of 100 (± 20) foot candles, and shall be a good match to the standard sample under incandescent lamplight at 2300 (± 200) Kelvin (see 4.4.2.4).

* 3.11.2 Colorfastness. The finished cloth shall show fastness to light, wet-dry cleaning, laundering, perspiration (acid test only) and crocking equal to or better than the standard sample. As a limit of acceptability, or when no standard sample is available, the finished cloth shall show a minimum of "good" fastness to light (40 standard fading hours), wet-dry cleaning, laundering, perspiration (acid test only), and crocking rating not lower than 4.0. Tests shall be as specified in 4.6.

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3.12 Finish. The type and character of the finish shall be equal to or better than the standard sample. The cloth shall be thoroughly scoured, properly fulled, (carbonized if necessary) face evenly sheared, and well pressed.

3.13 Wool content label. The cloth shall be labeled in accordance with the Wool Product Labeling Act of 1939 (see 2.1.2).

3.14 Marking. The back of the cloth shall be marked "BACK" at both ends of each roll with the letters not less than 1/2 inch (1.3cm) in height (see 4.4.2.3). The marking shall be clearly legible in any indelible marking medium used commercially and shall not show through on the face of the cloth.

3.15 Workmanship. The finished cloth shall conform to the quality established by this specification. The demerit points per 100 spare yards (83.6 Sq.m) when calculated as specified in Section 4 shall not exceed the established maximum point value. Each defect shall be marked with a 1 to 1-1/2 inch (2.5 to 3.8cm) long string. The string shall be sewn into the selvage near the defect. A red string shall represent a three or four point defect and a white string shall represent a one or two point defect (see 4.4.2.5).

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

* 4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

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1. First article inspection (4.3).
2. Quality conformance inspection (see 4.4).

* 4.3 First article inspection. When required (see 3.2 and 6.2), the first article shall be visually examined for color, finish, and appearance and shall be tested for chemical and physical properties in accordance with the methods specified in 4.6.

4.4 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-655, except where otherwise indicated.

4.4.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable procurement documents.

* 4.4.1.1 Wool grade. The wool shall be visually examined for grade in scoured form, prior to dyeing, picking, or carding by comparison with the applicable U.S. Standard (see 2.1.2). The examination for grade shall be performed on a composite sample of 1 pound (0.45 Kg) of scoured wool for each 10,000 pounds (4500 Kg) or fraction thereof in the lot. The composite sample shall be selected from not less than 10 percent of the bags/bales in the lot. In the event of a dispute resulting from above comparisons, the wool grade shall be determined by the width method (wedge). The lot shall be unacceptable if one or more composite samples fail to meet the requirements of 3.3.1. Each lot shall consist of one grade of wool only. The sample unit shall be 1 yard (0.91 m) of undyed top. The sample size (number of sample units) shall be as shown below. The lot size shall be expressed in units of 1 pound (.454 kg). The results from each sample unit shall be reported separately. The lot shall be unacceptable if a single sample unit fails to meet the required grade.

Lot size - pounds (kg)	Sample Size
800 (363) or less	2
801 (363) up to and including 22,000 (9988)	3
22,001 (9989) and over	5

4.4.1.2 In-process inspection. Inspection shall be made at any point, or during any phase, of the manufacturing process to assure that no laps, noils, or any other wool manufacturing by-products are used. The Government reserves the right to exclude from consideration for acceptance any material for which in-process inspection has indicated nonconformance.

4.4.2 Examination of the end item. Examination of the end item shall be in accordance with 4.4.2.1 through 4.4.2.5.

4.4.2.1 Yard-by-yard examination. The yard-by-yard examination shall be as specified in MIL-STD-655. The fabric quality shall be grade 10.

* 4.4.2.1.1 Limits of slubs and knots. Only those slubs and knots which exceed the limits shown on the Sears Fabric Defect Scale (see 6.4), D or 3-1/2 as applicable for slubs and C for knots shall be scored.

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4.4.2.2 Examination for length. The examination for length shall be as specified in MIL-STD-655.

4.4.2.3 Examination for back marking and compliance with the Wool Products Labeling Act. The examination for back marking and compliance with the Wool Products Labeling Act (see 2.1.2) shall be as specified in MIL-STD-655.

4.4.2.4 Examination for shade match and finish, individual rolls. Each roll in the lot shall be examined visually for shade match (see 3.11) and finish (see 3.12). A roll shall be unacceptable if it fails to meet the requirements for shade match or finish. The sample unit shall be a 4 inch by 20 inch (10.2 cm x 50.8 cm) swatch of the cloth. A sample unit shall be drawn from each roll in the lot.

4.4.2.5 Examination for defect marking (stringing of defects). The examination for defect marking shall be as specified in MIL-STD-655.

* 4.5 Examination of packaging requirements. Examination shall be made in accordance with the provisions of PPP-P-1132 to determine that put-up and preservation, packing, and marking comply with the Section 5 requirements.

4.6 Tests. The methods of testing specified in FED-STD-191, wherever applicable and as listed in Table II shall be followed. The physical and chemical values specified in Section 3 apply to the results of the determinations made on a sample unit for test purposes as specified in the appplicable test methods. The sample unit shall be 3-1/2 continuous yards (3.19 continuous meters) full width f the finished cloth. The lot shall be unacceptable if one or more sample units fail to meet any test requirements specified. The lot size shall be expressed in units of 1 yard (0.91 m) each. All test reports shall contain the individual values utilized in expressing the final result. The sample size (number of sample units) shall be in accordance with the following:

Lot size-yards (meters)	Sample size
800 (728) or less	2
801 (729) up to and including 22,000 (20,020)	3
22,001 (20,021) and over	5

Table II. Test methods

Characteristic	Requirement paragraph	Test Method
Yarn ply	3.3.2	Visual
Weave	3.4	Visual 2/
Weight	3.5	5041
Yarns per inch:		
Warp	3.5	5050
Filling	3.5	5050

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Characteristic	Requirement paragraph	Test Method
Breading strength:		
Warp	3.5	5100
Filling	3.5	5100
Shrinkage	3.5	5590
Wool content	3.7	2102 1/
pH	3.8	2811
Mothproofing	3.10	3/
Colorfastness:		
Light	3.11.2	5660 4/
Wet-dry cleaning	3.11.2	5622
Perspiration	3.11.2	5680
Crocking	3.11.2	AATCC-TM-8-1988
Laundering	3.11.2	5614 5/

1/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirements.

2/ One determination per sample unit and the results reported as "pass" or "fail".

3/ As tested in MIL-C-43665.

4/ The calibration of the fadeometer and the definition of AATCC Fading Units shall be according to AATCC Test Method 16-1987. Using the formula below, 16 to 22 AFU's are acceptable as 20 SFH's.

$$\text{AATCC Fading Units} = \frac{(\Delta E^*)\ (20)}{1.7}$$

Where (ΔE^*) is the color difference in CIELAB* units of color difference.

5/ Only the stain on the wool fibers of the color transfer cloth shall be evaluated.

5. PACKAGING

5.1 Put-up and preservation. Put-up and preservation shall be level A or Commercial as specified (see 6.2).

* 5.1.1 Level A and Commercial. The cloth shall be put up and preserved in accordance with the applicable requirements of PPP-P-1132.

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5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

5.2.1 Levels A, B, or Commercial. The cloth shall be packed in accordance with the applicable requirements of PPP-P-1132.

5.3 Marking. In addition to any special marking required in the contract or order, shipments shall be marked in accordance with the requirements of PPP-P-1132.

* 5.4 MIL-STD-2073. When specified (see 6.2), Preservation and Packing shall be in accordance with MIL-STD-2073-1 and MIL-STD-2073-2.

6. NOTES

*(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory).

6.1 Intended use. The cloth covered by this specification is intended for use in the manufacture of the Navy men's and women's clothing.

* 6.2 Acquisition requirements. Acquisition document should specify the following:

a. Title, number, and date of this specification.
b. Type of cloth required (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When first article sample is required (see 3.2). The item will be tested and should be a first article sample. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, quantity, testing and approval of the first article.
e. Width required if other than specified (see 3.6).
f. Length required if other than specified (see 3.9).
g. Levels of put-up, preservation, and packing (see 5.1 and 5.2).
h. When all packaging data is to be found in MIL-STD-2073-1 and MIL-STD-2073-2 (see 5.4).

6.3 Standard sample. For acess to standard sample, address the procuring office issuing the invitation for bids (see 3.1).

* 6.4 Sears Fabric Defect Scales are available from Sears, Roebuck and Company, Department 817 (Attn: BSC 23-29), "Sears Defect Replica Kit", Sears Tower, Chicago, IL 60684.

* 6.5 Recycled material. The use of recycled material is not permitted.

6.6 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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* 6.8 Subject term (key word) listing.

Fabric, Navy clothing
Finish, Moth-proofed
Yarn, wool

Custodian:
Navy - NU

Preparing Activity:
Navy - NU

Review Activity:
DLS - CT

Project No. 8305-N383

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions -- Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-16290K(NU)	CLOTH, MELTON, WOOL (MOTH-PROOFED)

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* -- Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* -- Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* -- Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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